

Sunlight: make a safer plan

Teacher guide and worked answers

Explain why sunlight can harm eyes and choose sensible ways to reduce exposure.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

[_passsb/inputs/Build SOW 2026-2027.xlsx](#) | BUILD Weekly - Spring | C35

Explain that the Sun's light can be dangerous and how to protect our eyes (links PSHE).

Direct match to the named outcome. The lesson is a 40-minute exemplar, with optional greater depth; no accreditation completion is claimed.

Prior learning

Everyday objects can be described by what we observe.

Use a labelled diagram or evidence card to explain a choice.

Success evidence

Explain that sunlight can damage eyes.

Choose suitable UV-filtering sunglasses, a brimmed hat and shade.

Keep the rule: never look directly at the Sun.

Equipment

Printed scenario and equipment cards; optional empty sunglasses packaging with clear UV information; paper and pens.

Preparation

Use the supplied fictional cards, not pupils' personal eyewear, for judgments.

Read the linked NHS/RNIB guidance. Follow the school's normal outdoor arrangements and individual pupil plans.

Practical care

This is an indoor planning lesson: never look directly at the Sun or stare into bright reflections.

Never test protection by looking at the Sun through lenses, plastic, a camera, binoculars or a telescope.

Do not use UV lamps or test products on pupils. Sunglasses are for ordinary outdoor protection, not Sun viewing.

Ready to teach alternative

The full lesson is a paper-based decision task. No outdoor exposure, equipment testing or online video is required.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Plan an outdoor break
2	Retrieval	2-5	Three light clues
3	I do	5-9	Some sunlight is invisible
4	I do	9-12	I read the protection evidence
5	I do	Within stage	Reflection matters as well
6	We do	12-16	Choose the strongest kit
7	We do	16-20	Repair the risky message
8	Hinge	20-23	Which is the safest science claim?
9	You do	23-25	Choose your investigation route
10	You do	Within stage	Use the evidence
11	You do	25-35	Make your own science record
12	You do	Within stage	Check and improve
13	Review	35-38	Lundy loop: improve our account
14	Exit	38-40	Three quick checks

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Plan an outdoor break

State clearly that this is an indoor planning activity. The lesson is about sensible choices, not testing limits of exposure.

2 Retrieval Three light clues

Answers yes, yes, no. Correct the third immediately; no pupil should demonstrate or rehearse unsafe Sun viewing.

3 I do Some sunlight is invisible

Think aloud: "I cannot judge UV by how bright or hot I feel. This diagram separates two kinds of radiation; the boxes do not show their amounts. I choose protection before exposure." Do not imply UV is the only mechanism of harm from direct Sun viewing.

4 I do I read the protection evidence

Think aloud: "A dark-looking lens changes visible light. I need reliable information about UV filtering and suitable fit. In real life an adult checks the product through a reputable supplier or optician." Ordinary sunglasses never make staring at the Sun safe.

5 I do Reflection matters as well

Shares I do time. Think aloud: "Facing away is not a complete protection plan. Light can arrive from another direction. I still use suitable sunglasses, a hat and shade." This is a path model, not a measured dose estimate.

6 We do Choose the strongest kit

Pairs receive kit cards A-D. A has only a dark tint claim; B is a constructed verified suitable UV-filtering option. Select B and add C/D. Do not teach pupils to trust any unverified sticker in the real world.

Reveal: Choose B with a hat and shade. A's darkness alone is not enough.

7 We do Repair the risky message

Ask every pair to reject the message and provide a replacement. Give a quiet written route. Then introduce scenario E by water and F with only dark unverified glasses: neither licenses direct viewing or guessing protection.

Reveal: Never look directly at the Sun. Suitable sunglasses reduce ordinary exposure; they do not make Sun viewing safe.

8 Hinge Which is the safest science claim?

If pupils choose A, distinguish visible tint from UV filtering. If B, restate the direct-viewing prohibition before moving on.

A. Very dark lenses prove good UV protection. - Tint alone does not confirm UV filtering; reliable product information is needed.

B. Sunglasses let us stare at the Sun safely. - Ordinary sunglasses do not make direct Sun viewing safe.

C. Use suitable UV protection, a hat and shade; never look directly at the Sun. - This combines sensible protection and preserves the essential no-looking rule.

9 You do Choose your investigation route

Give one route only, selected with the pupil. The shared evidence page supports every route. The independent task is a decision brief, not a real exposure trial. An adult can scribe the pupil's reasons. Use the supplied verified fictional product information; never judge a real pupil's glasses by colour alone.

10 You do Use the evidence

Shares the 12-minute independent stage. The independent task is a decision brief, not a real exposure trial. An adult can scribe the pupil's reasons. Use the supplied verified fictional product information; never judge a real pupil's glasses by colour alone.

11 You do Make your own science record

Allow ten minutes of independent response after the two-minute setup. The independent task is a decision brief, not a real exposure trial. An adult can scribe the pupil's reasons. Use the supplied verified fictional product information; never judge a real pupil's glasses by colour alone.

12 You do Check and improve

Shares the independent stage. Ask the pupil to point to the clue supporting their answer; accept a spoken correction or changed choice.

13 Review Lundy loop: improve our account

Give a private or paired option. Adopt one scientifically sound pupil revision in the class explanation and name the change. No public ranking.

Reveal: Never look directly at the Sun, even when wearing ordinary sunglasses.

14 Exit Three quick checks

Collect brief individual answers. Secure core requires the main idea and a relevant reason. Revisit the relevant worked diagram next lesson if the explanation reverses cause and effect.

Reveal: See the teacher answer guide for the three checks.

Answers and assessment

Retrieval 1-3

1 yes. 2 yes. 3 no; never look directly at the Sun.

We do W1-W2

W1 B has the relevant UV and fit evidence; add C and D. A is unconfirmed. W2 never look directly at the Sun. Suitable sunglasses are for ordinary exposure, not staring at it.

Hinge

Third option correct. Dark tint does not itself confirm UV protection, and ordinary sunglasses do not permit Sun viewing.

S1-S5

S1 B. S2 wide-brimmed hat and shade. S3 no. S4 never. S5 darkness describes visible light transmission, not confirmed UV filtering.

T1-T5

T1 direct sunlight and reflected sunlight from water. T2 B because reliable evidence confirms suitable UV protection and fit. T3 hat and shade. T4 never look directly at the Sun. T5 exposure is reduced, but ordinary sunglasses do not make direct viewing safe and protection is not an unlimited guarantee.

X1-X4

X1 reliable UV protection and suitable fit information. X2 use the supervised indoor option and ask an adult to plan suitable protection before going out; do not improvise a Sun test. X3 check reputable product information or consult an optician; an adult should also check fit. X4 UV is invisible, so perceived brightness alone is insufficient.

Exit E1-E3

E1 sunlight can harm eyes. E2 any two: suitable UV-filtering sunglasses, wide-brimmed hat, shade. E3 no.

R1-R2

Responses vary. Credit a science-related revision or relevant next question. A pass or private response is acceptable; preferences have no fixed right answer.

Teaching and assessment notes

40 minutes: opening 2; retrieval 3; I do 7; We do 8; hinge 3; You do 12; review 3; exit 2. Zero-minute slides share the stage time.

BUILD follows the user's Year 3 science anchor with age-respectful materials for secondary pupils. Routes are adjustable access choices, not fixed groups or diagnoses.

Every core task can run from the printed evidence. Do not present constructed teaching evidence as observations pupils made. A pupil's spoken reasoning carries the same scientific value as writing.

Keep advice proportionate and practical. No frightening images or personal medical stories are required. The lesson teaches ordinary prevention, not diagnosis or a product certification test.

The NHS and RNIB guidance provides the public-health basis. The diagrams simplify categories and pathways; they do not quantify an individual pupil's exposure or show a complete electromagnetic spectrum.

HTML We do: pupils choose a prediction, test it against the user-controlled SVG model or supplied evidence, then explain or revise. The shared paper cases and W1/W2 prompts provide the equivalent classroom discussion without a device. Any animation is a teaching model, not filmed experimental evidence.

Access and responsive teaching

Supported

Choose one route: short choices, word bank and pointing or adult-scribed explanation; keep the same science goal.

Standard

Use the evidence provided to record a decision and explain the science in short sentences.

Stretch

Apply the same core idea to a changed setup, identify a limitation and justify a better decision.

Regulation

Preview the sequence and equipment. Offer seated observation, a quiet paper route, partner rehearsal and a pass. Never require public performance or a darkened room.

Misconceptions to check

Dark lenses must block UV.

Darkness shows reduced visible light; it does not by itself confirm UV protection.

Check: Ask which card provides UV evidence.

Sunglasses make looking directly at the Sun safe.

Ordinary sunglasses do not make direct Sun viewing safe.

Check: Ask for the rule that always remains.

Only direct sunlight matters.

Reflected sunlight can also reach us; combine suitable protection.

Check: Use the water reflection diagram.

Word	Meaning
ultraviolet (UV)	Radiation in sunlight that we cannot see and that can harm eyes and skin.
protection	Something that reduces exposure or harm.
reflect	To send radiation back from a surface.

Scientific references

NHS: [sunscreen and sun safety](#)

Checked 8 September 2026. Eye protection, suitable sunglasses, hats, shade and never looking directly at the Sun.

RNIB: protect your eyes from the sun

Checked 8 September 2026. UV-filtering sunglasses and combined protection; children should never look directly at the Sun.

Met Office: how UV can affect your eyes

Checked 8 September 2026. Suitable UV-protective sunglasses and shade; distinguishes invisible UV exposure from visible brightness.

Department for Education: science programmes of study

Checked 8 September 2026. Year 3 light outcomes anchor the BUILD content; ray explanations are carefully modelled as an accessible extension.